The Role of Public Participation and Governance in Designing Spaces Under Flyovers: A Study of Commoning and Gender-Inclusive Public Spaces

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Key Words: Public Space, Commoning, Governance, Gender Inclusive, Spaces under flyover

1. Introduction

Urban flyovers have always been relied on for managing vehicular traffic in rapidly urbanizing cities. However, the spaces under these structures have been neglected and underutilized in the past. These spaces have caught the attention recently but remain dominated by informal uses such as informal parking, waste dumping or are left abandoned. These underutilized spaces are ubiquitous in urban context yet they lack coherent design which leads them to be treated as residual spaces with limited functionality. These areas, however hold immense potential for transformation into vibrant urban commons that foster social interaction, community needs, economic activities and cultural expression. With thoughtful interventions they hold the potential to become inclusive environment that could cater to the diverse requirement of the society.

Inclusivity refers to creating spaces that are accessible to all individuals regardless of their diversity in various categories or background. It ensures that public spaces foster equity, safety, and a sense of belonging thereby allowing diverse individual and community needs to coexist and thrive. Elimination of physical and social barriers may ensure the sense of commoning among public. All these transformation to urban commons requires multi-dimensional approach. Effective governance ensures sustainability, active public participation ensures community ownership, inclusivity and gender sensitive design principles ensure these spaces are safe and accessible to all. Together, these provide a foundation for commoning, a process that emphasizes collective ownership, shared management, and equitable access to public spaces.

2. Conceptualizing the topic through literature review

- The democratic process of city making should be open to transformation from the bottom up, engaging citizens in the initiation, design and development of public space. (Bagni et al., 2017)
- Effective public participation – meaningfully involving the broad public in issues that matter – is key to good governance (Menon and Hartz-Karp, 2019)
- Public participation also begins laying the ground work for sustainable practices in physical planning and management as well as social community building. (Loures and Crawford, 2008)
- Commoning practices can thus play a transformative role in counteracting the loss of access to urban space as a result of urban regeneration and densification (Bresnihan & Byrne, 2015)
- Thus, a shift occurred from understanding commons as a resource to be managed as efficiently as possible, to commons as commoning – a process of building community and developing alternative forms of resource management, acting against the influence

of capitalist markets and for-profit development over local resources. (Verheij et al., 2024)

- Engaging with public space often means dealing with hostile architecture, short-sighted urban policies, and urban planning that imagines a 'neutral user', 'neutral' in terms of gender, race or caste, physical ability, and sexuality, which inevitably caters to already-privileged groups (al., 2024)

3. Key words

- **Public Space**: Public spaces must be seen as multi-functional areas for social interaction, economic exchange, and cultural expression among a wide diversity of people.
- **Commoning**: Urban commons are shared resources in the city that are managed by their users in a collaborative and non-profit-oriented way.
- **Governance**: Governance refers to the systems, policies, and processes through which decisions about public spaces are made and implemented
- **Gender Inclusive**: A gender-inclusive approach ensures that public spaces are accessible, safe, and accommodating for all genders
- **Spaces under flyover**: Spaces under flyovers are often underutilized and neglected urban areas with significant potential for transformation into vibrant public spaces.

4. Theoretical Mapping

This paper shall establish a structured framework for analysing the concepts and relationships integral to the research topic, focus on key theories and explore their interconnectedness.

4.1 Identify the concepts

- Commoning
- Urban Commons
- Public participation
- Governance
- Gender-inclusivity

These concepts represent an important dimension of the study addressing various aspects that contribute to the transforming of spaces under flyovers into inclusive spaces

4.2 Defining the concepts

- **Commoning**: The social practices of creating and managing shared resources collaboratively and taking collective ownership.
- **Urban Commons**: The shared public spaces that function as multi use hubs for diverse activities.
- **Public Participation**: Mechanisms through which communities influence urban planning and urban design processes. Ensures that diverse opinions, voices are heard in shaping the urban spaces.
- **Governance**: Frameworks and policies that facilitate or hinder participatory processes
- **Gender- Inclusivity**: Ensuring equal representation and addressing gender-specific needs in public space design.

4.3 Operational Definitions

- Public participation involves activities such as surveys, workshops, and public consultations.
- Governance mechanisms include policies, local authority interventions, and partnerships with NGOs.
- Commoning includes shared activities such as markets, art installations, or recreational zones.

4.4 Variables

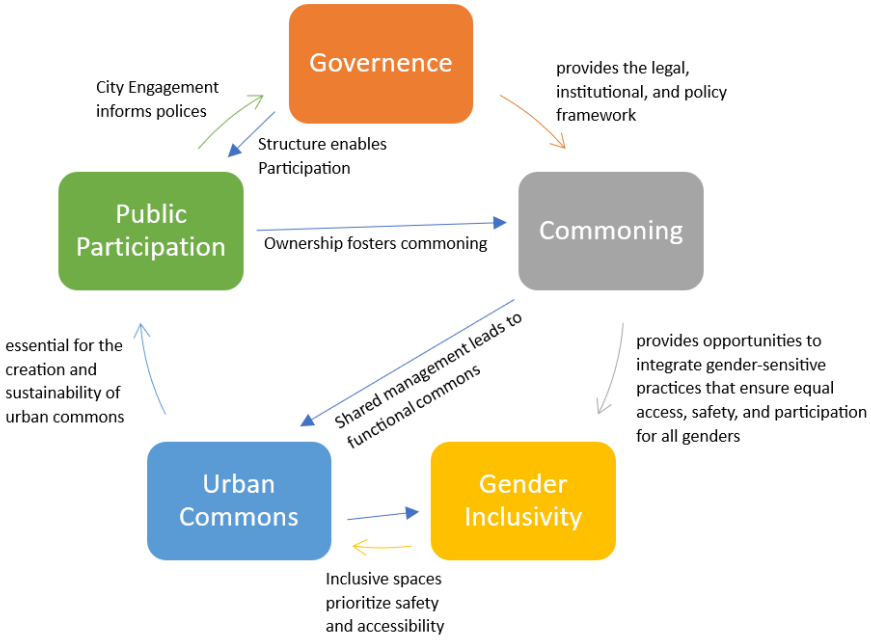
- **Independent Variables:**
  - Types of participatory mechanisms employed (e.g., surveys, focus groups).
  - Governance structures (e.g., municipal policies, funding, stakeholder engagement).
  - Design interventions (e.g., lighting, activity zones).
- **Dependent Variables:**
  - Inclusivity and safety of the space.
  - Participation levels of diverse community groups.
  - Functionality and user satisfaction.

4.5 Interconnectedness

- Public Participation ↔ Governance:
  - Public input strengthens governance by identifying community needs, while governance frameworks enable structured participation.
- Public Participation ↔ Commoning:
  - Community engagement fosters shared ownership and responsibility for spaces, making commoning viable.
- Governance ↔ Gender Inclusivity:
  - Gender-sensitive governance ensures the inclusion of marginalized groups in both design and decision-making processes.
- Commoning ↔ Urban Commons:
  - Shared ownership transforms underutilized spaces into functional urban commons that reflect community values.
- Urban Commons ↔ Gender Inclusivity:
  - Inclusive urban commons incorporate design elements that prioritize safety, accessibility, and representation for all genders.

5. Conceptual Framework

The conceptual framework of this research establishes the relationships between key concepts which include public participation, governance, commoning, urban commons, and gender inclusivity. These concepts drive the transformation of under-flyover spaces into inclusive urban environments. It provides a structured approach to understanding how governance mechanisms and participatory practices shape the design and functionality of these spaces while ensuring gender equity.



To address the research objectives and question, the above illustrated conceptual framework will be employed. Governance structures enable public participation by establishing legal frameworks, participatory platforms, and financial support, ensuring that citizens actively shape urban spaces. In turn, public participation informs governance, making policies more inclusive and reflective of community needs. Governance also plays a vital role in facilitating commoning by providing legal recognition and resources for community-led space management. Commoning fosters gender inclusivity by integrating women and marginalized groups into shared governance, ensuring safer, more accessible urban spaces. Urban commons and public participation are mutually reinforcing—public engagement transforms underutilized flyover spaces into urban commons, while commons serve as platforms for ongoing civic participation. Together, these interconnections create sustainable, inclusive, and community-driven urban spaces under flyovers, preventing exclusionary development and ensuring long-term usability.

6. Theoretical Framework

The theoretical Framework will connect key concepts of public participation, governance, gender inclusivity etc in public spaces under flyovers. The list of the theories that can be incorporated are:

a. **Environmental Psychology** - Explores the relationship between individuals and their physical surroundings. It helps understand how urban spaces under flyovers influence user behaviour, comfort, and safety.

b. **Urban Commons Theory** - Emphasizes spaces as shared resources governed by collective action and shared responsibility.

c. **Gender and Space Theory** - Examines how spatial design and planning perpetuate gender inequalities by shaping who can access, use, and feel comfortable in public spaces.

d. **Placemaking Theory** - Emphasizes community-led processes to create spaces that are functional, inclusive, and representative of the users' needs.

Environmental Psychology examines the relationship between individuals and their physical surroundings, highlighting how the design of urban spaces influences user behaviour, comfort, and safety. In this context, it helps analyse the psychological impact of lighting, accessibility, and spatial organization on the usability of under-flyover spaces. Urban Commons Theory emphasizes the concept of shared spaces managed collectively by communities through collaborative actions and shared responsibilities. This theory is applied to explore how such neglected spaces can be reimagined as urban commons through participatory governance, fostering a sense of collective ownership and equitable use. Gender and Space Theory critiques traditional urban planning practices that perpetuate gender inequalities by limiting accessibility and comfort for women and marginalized genders. Its application focuses on identifying and addressing the spatial barriers that hinder inclusivity, proposing gender-sensitive designs such as safe pathways, lighting, and gender-neutral facilities. Finally, Placemaking Theory highlights the importance of community-led processes in creating spaces that reflect the needs and identities of their users. It emphasizes participatory design as a tool to transform underutilized areas into vibrant, multifunctional spaces that encourage social interaction and community engagement. Together, these theories offer a comprehensive lens to evaluate and address the social, psychological, and structural challenges in the transformation of under-flyover spaces, ensuring they evolve into inclusive, accessible, and sustainable urban commons.

7. Research Design

7.1 Objectives of the Study:

- To understand the role of public participation and how it influences the design and use of public spaces.
- To analyse public participation mechanisms and their effectiveness in achieving gender-equal representation and inclusivity in such designs.
- To analyse how the local/urban governance facilitate or hinder public participation in public spaces.
- To understand the role of governance in transforming under-flyover spaces into urban commons.
- To investigate existing urban interventions under flyovers, assessing their condition in terms of governance, public participation, and the inclusion of gender-sensitive design elements in the usage and functionality of these spaces.
- To analyse how under-flyover spaces in the city facilitate commoning practices, identifying factors that promote or inhibit communal activities in these areas.

7.2 Research Questions:

- How do government policies affect the design and management of public spaces with a focus on spaces under flyovers?

- How does public participation influence the design of public spaces with a focus on spaces under flyovers?
- How do existing spaces under flyovers function as urban commons, and what factors promote or inhibit commoning practices?
- What is the role of public participation in transforming these spaces into urban commons?
- How can gender equity be integrated into public participation and governance frameworks to ensure inclusive design for spaces under flyovers?

7.3 Methodology:

The study has adopted a mixed-methods approach to explore both the tangible and intangible factors influencing the utilization and functionality of public spaces under flyovers.

- Qualitative Methods
  - Conduct interviews with urban planners, policymakers, and community leaders to understand the challenges in governance and public participation.
  - Engage with users to understand gender-specific safety concerns, accessibility issues, and desired amenities.
  - Analyse successful interventions secondary case studies. (Matunga Flyover Garden in Mumbai,
  - Visit existing under-flyover spaces to assess the current condition as primary case study
- Quantitative Methods
  - Conduct surveys to users of under-flyover spaces to collect numerical data on gender-specific experiences and preferences.
  - Use GIS tools to analyse how under-flyover spaces are distributed and utilized.
  - Review existing datasets (e.g., census, urban policy data, or crime statistics) to identify gender disparities in urban space usage

7.4 Case Study:

7.4.1 Primary Case Study

7.4.1.1 Kathipara urban square, Chennai

7.4.1.1.1 Introduction

Kathipara Urban Square, located under the Kathipara Flyover in Chennai, is a revolutionary example of how a neglected under-flyover space can be transformed into vibrant urban commons. Previously an unutilized and semi - transit zone, the area has been reimagined and designed as a user - friendly urban plaza, enhancing its accessibility, usability, and public engagement. Currently the plaza is in its full potential even at late night hours. This in turn generates constant movement in and around the urban plaza making the area safer for public. Developed by the Chennai Metropolitan Development Authority (CMDA), this project has projected how effective governance,

public participation, and gender-inclusive planning can redefine urban spaces to serve diverse community needs.

7.4.1.1.2 Need for Intervention

Before its transformation, the space under Kathipara Flyover was:

- Underutilized and partially accessible. A portion of the area had a functional children's park but did not engage to its full potential initially.
- No dedicated crossings or public amenities making it unsafe for the public.
- Neglected, lacking public infrastructure, proper landscape or organized space utilization.

Recognizing the potential of under-flyover spaces, the CMDA initiated the Urban Square project to repurpose this area into a multi-functional, inclusive, and pedestrian-friendly environment.

7.4.1.1.3 Design and Development

The transformation of Kathipara Urban Square focused on four key urban design principles:

1. Public Participation & Engagement
    - Community consultations helped identify essential public amenities.
    - Feedback was incorporated into the design process, ensuring that the space met local needs.
2. Pedestrian-Centric Design
    - Dedicated walkways, seating zones, and landscaping made the space accessible and comfortable.
    - Traffic-calming measures ensured safe pedestrian movement.
    - Dedicated parking areas to avoid vehicular movement inside the core area of the plaza
3. Functional & Multi-Use Space
    - The square features seating areas, food kiosks, exhibition zones, and recreational spaces in one part of the square.
    - The open plaza allows for local handicraft markets and community gatherings.
4. Gender-Inclusive & Safe Public Space
    - Adequate lighting, security features, and visibility enhancements addressed safety concerns.
    - The design accommodates women, children, elderly, and differently-abled individuals, ensuring equitable access.

7.4.1.1.4 Governance and Policy Support

Chennai Metropolitan Development Authority (CMDA) played a central role in planning and funding. Collaboration with municipal bodies, traffic authorities, and local organizations ensured integrated space management. Policy frameworks promoted the adaptive reuse of under-flyover spaces, setting a precedent for similar projects.

7.4.1.1.5 Impact and Success

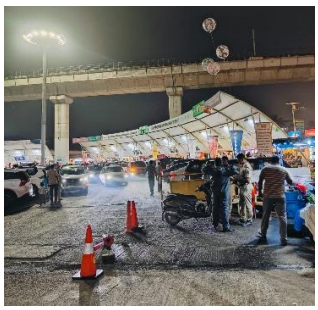
Kathipara Urban Square has significantly improved the quality of public space in Chennai:

- **Increased Footfall**: The plaza has become a popular destination for daily commuters, pedestrians, and local vendors.
- **Enhanced Safety**: Women and marginalized groups report feeling safer due to improved visibility and active space usage.
- **Economic Opportunities**: The inclusion of small-scale vendors, local artisans, and food stalls has created economic benefits.
- **Urban Aesthetics**: The space now functions as a green urban oasis, incorporating trees, seating areas, and recreational zones.

7.4.1.1.6 Conclusion

Similar models can be implemented under flyovers in other Indian cities to address underutilized spaces. Periodic maintenance, public engagement, and governance oversight are necessary to retain usability. Additional facilities, such as bicycle parking, digital kiosks, and shaded seating, could enhance the experience further. Kathipara Urban Square is a benchmark project in Chennai's urban development landscape, showcasing how strategic governance, participatory planning, and inclusive design can transform underutilized spaces into thriving public assets. This case study reinforces the importance of reclaiming urban voids for collective use, ensuring accessibility, safety, and long-term community benefits.



7.4.1.2 New town Bridge – Biswa Bangla Sarani, Kolkata

7.4.1.2.1 Introduction

The New Town Bridge at Biswa Bangla Sarani in Kolkata has been transformed into a vibrant and functional public space that fosters community engagement. Unlike many underutilized or neglected spaces beneath flyovers, this area has been creatively adapted to serve as an inclusive, active urban commons. The integration of local crafts, seating areas, and artistic interventions has contributed to its success as a shared community space.

7.4.1.2.2 Spatial Organization

The space under the New Town Bridge is structured into two primary functional zones:

- **Arts & Crafts Marketplace:** One section is dedicated to local artisans who regularly set up stalls to sell handmade products. These include traditional Bengali crafts, textiles, pottery, and other indigenous art forms, creating an economic and cultural hub.
- **Community Kiosks & Seating Spaces:** The other section provides seating areas and kiosks that support informal gatherings and small-scale commercial activities. This side promotes social interaction and offers respite for pedestrians.
- **Murals & Public Art:** The bridge's underside is adorned with large, colourful geometric murals, enhancing the space's aesthetic appeal. These artistic elements contribute to placemaking by creating a visually engaging environment that celebrates Kolkata's creative spirit.

7.4.1.2.3 Governance & Public Participation

The activation of this under-flyover space has been driven by proactive governance and local participation:

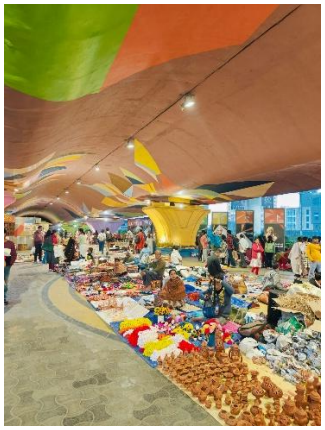
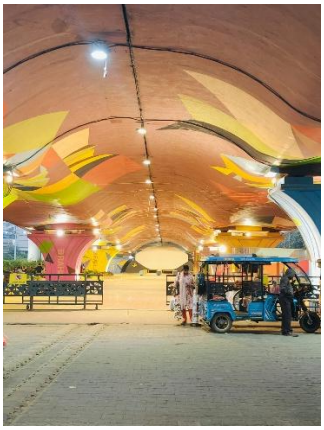
- **Municipal & Institutional Support:** The local government has played a critical role in maintaining the space, allowing for structured vendor operations and periodic events.
- **Community Engagement:** Regular activities, craft fairs, and cultural programs ensure that the space remains dynamic and accessible to a broad demographic.

7.4.1.2.4 Urban Commons & Gender Inclusivity

- **Commoning Practices:** The area functions as an urban commons where vendors, pedestrians, and visitors interact, creating a self-sustaining public domain.
- **Gender-Inclusive Design:** The presence of well-lit areas, continuous activity, and designated seating zones enhances safety, making the space more inclusive for women and marginalized groups.

7.4.1.2.5 Conclusion

The New Town Bridge – Biswa Bangla Sarani showcases the successful transformation of an underutilized space into a thriving urban commons through strategic design, governance, and community participation. By integrating local crafts, public seating, and artistic interventions, the space not only supports small-scale economic activities but also fosters a sense of belonging among diverse user groups. The presence of well-lit and vibrant public art enhances safety, encouraging greater inclusivity, particularly for women and marginalized communities. Furthermore, this initiative highlights the importance of governance structures in enabling public participation, demonstrating how municipal support and grassroots engagement can co-create dynamic public spaces. The space also exemplifies the principles of commoning, where shared urban infrastructure is repurposed to meet social, cultural, and economic needs. As cities continue to grapple with space constraints, the New Town Bridge serves as a replicable model for reimagining flyovers as multifunctional public assets, promoting sustainable urbanism and inclusive placemaking.


7.4.2 Secondary Case Study

7.4.2.1 Domlur Flyover, Bengaluru.

7.4.2.1.1 Introduction

The area beneath the Domlur Flyover in Bengaluru has undergone a significant transformation, evolving from a neglected space into a vibrant community hub. This revitalization showcases the potential of urban spaces when innovative design and community engagement are prioritized.

7.4.2.1.2 The Need for Intervention

Prior to its redevelopment, the under-flyover space at Domlur was underutilized and often misused, leading to concerns about safety and urban blight. Recognizing the potential of this area, the Bruhat Bengaluru

Mahanagara Palike (BBMP) initiated a project to convert the space into a functional public area.

7.4.2.1.3 Design and Development

The transformation process involved several key steps:

1. Conceptualization of an Open-Air Theatre: BBMP proposed the construction of an open-air theatre beneath the Domlur Flyover to provide a cultural and recreational space for the community.
2. Community Engagement: Residents and organizations were involved in the planning process to ensure the space met the community's needs and preferences.
3. Implementation: Construction of the open-air theatre commenced with an estimated budget of ₹40 lakh, aiming to create a venue for cultural events, performances, and community gatherings.

7.4.2.1.4 Impact and Successes

The redevelopment of the space under the Domlur Flyover has led to several positive outcomes:

- Cultural Enrichment: The open-air theatre has become a venue for cultural events, performances, and community gatherings, enriching the local cultural landscape.
- Community Engagement: The project has fostered a sense of ownership and pride among residents, leading to increased community participation in maintaining and utilizing the space.
- Urban Aesthetics: The transformation has improved the visual appeal of the area, contributing to the overall aesthetics of the neighbourhood.

7.4.2.1.5 Conclusion

The Domlur Flyover project offers valuable insights:

- Innovative Use of Space: Transforming under-flyover areas into functional spaces like open-air theatres can address urban challenges and meet community needs.
- Collaborative Planning: Engaging the community in the planning and development process ensures the space serves its intended purpose and fosters a sense of ownership.
- Sustainable Maintenance: Ongoing community involvement is crucial for the continued success and upkeep of such public spaces.

The transformation of the space beneath the Domlur Flyover into an open-air theatre demonstrates the potential of reimagining underutilized urban areas. This initiative highlights the importance of innovative design, community engagement, and collaborative governance in creating vibrant public spaces that enhance the urban experience.

8. Scope and Limitations

• **Scope**

• To focus on utilization of spaces under flyovers as public spaces, emphasizing the role of public participation and governance in their design and functionality.
• To highlight the gendered dimension of public spaces and to identify challenges and opportunities for achieving inclusivity.
• To Exploring the role of commoning practices in transforming under-flyover spaces into vibrant, community-driven urban commons.

• **Limitations**
• The study only considers generic genders (male and female), and does not explore non-binary or other gender identities due to time constraints and limitations in available data.
• Findings from specific flyovers and cities may not be generalized to all urban flyovers due to unique local contexts

9. Conclusion

The study of spaces under flyovers highlights their transformative potential in urban environments, where they can evolve from neglected, unsafe zones into vibrant public areas. Through a comparative analysis of various case studies, it is evident that such interventions require a blend of thoughtful design, community engagement, and strong governance to be successful. One of the key takeaways is that under-flyover spaces, when designed with inclusivity and accessibility in mind, can cater to diverse user groups, including pedestrians, cyclists, street vendors, and cultural performers.
A critical component of these transformations is the role of public participation. Cities that involve local communities in decision-making processes tend to create more resilient and sustainable urban spaces. Additionally, urban governance plays a crucial role in maintaining these spaces post-intervention, ensuring their upkeep and continued relevance. Without adequate governance structures, these revitalized spaces risk reverting to their previous states of neglect. These spaces when thoughtfully reimagined, contribute to the broader concept of urban commons, where public spaces are collectively created, managed, and utilized by local communities. The Kathipara Urban Square in Chennai exemplifies how well-designed infrastructure can provide a platform for diverse social interactions, from pedestrian movement to recreational and commercial activities. Similarly, the New Town Bridge in Kolkata fosters a strong sense of communing by enabling local craftsmen and vendors to showcase handmade products, transforming the space into a cultural and economic hub. These cases highlight that under-flyover spaces can transcend their infrastructural function and emerge as community-driven urban commons, reinforcing local identities and strengthening urban resilience.
Furthermore, a gender-sensitive approach is crucial in designing under-flyover spaces. Ensuring well-lit, pedestrian-friendly environments with designated areas for diverse activities can improve safety and encourage wider community engagement. The case studies illustrate that urban interventions should not only focus on aesthetics but also prioritize social inclusivity and sustainability.
Moving forward, city planners and policymakers must recognize the latent potential of underutilized spaces in cities. The study highlights that a multidisciplinary approach, integrating urban design, governance, and community participation, is the key to unlocking the full potential of these spaces.

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